
Scaling Laws for Language Models on SVG Code

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Abstract

Neural scaling laws describe power-law relationships between model size and loss, but their applicability to structured, non-linguistic domains remains largely unexplored. We investigate whether these relationships hold for autoregressive SVG code generation, training five decoder-only Transformer models spanning five size classes (1.3M–88.1M parameters under SP; 1.8M–91.2M under μ P, which unties input embedding and output projection weights) for one epoch on a 107M-token SVG corpus under both standard parameterization (SP) and Maximal Update Parameterization (μ P).

Under SP, the Tiny-optimal learning rate causes Large and XL models to diverge, preventing a meaningful power-law fit. Under μ P, the same rate transfers without adjustment across all five sizes, and the resulting losses follow a power-law trend ($L = 0.85 \cdot N^{-0.043}$, $R^2 = 0.76$). Large standard errors ($\alpha = 0.043 \pm 0.715$, $c \approx 0$, $c_{\text{err}} = 7.10$) reflect the constraint of fitting three parameters to five points, and preclude precise quantification or statistical comparison with NLP scaling exponents.

Despite loss generally decreasing with scale, generation quality appears highest at Medium (13.8M parameters, 90% unconditional render rate; $n=10$ samples per size) while models at other sizes score 60–70% (CIs overlap with Medium’s [60–98%])—a non-monotonic pattern consistent with a Chinchilla data-limited regime at larger scales. Our results suggest that scaling laws may predict loss trends but not downstream output quality in data-limited structured domains.

Code available at: <https://github.com/katsukii/svg-scaling-project>

1 Introduction

1.1 Motivation

Neural scaling laws describe empirical power-law relationships between model size and loss, characterized empirically by Kaplan et al. [2020] for natural language. Hoffmann et al. [2022] subsequently revised Kaplan’s compute-optimal allocation, showing that data and model size should be scaled equally rather than prioritizing model size. These regularities have guided compute allocation and architecture selection in NLP, but their applicability to structured, non-linguistic domains remains largely unexplored. Scalable Vector Graphics (SVG) presents an interesting test case: SVG code is XML-based text with deterministic syntax rules, a constrained vocabulary, and the unusual property that its outputs can be rendered visually, enabling evaluation beyond perplexity alone.

This work investigates three research questions in the context of autoregressive SVG code generation. **Q1:** Does the power-law relationship $L = a \cdot N^{-\alpha} + c$ hold for SVG? **Q2:** How do the fitted scaling exponent α and irreducible loss c compare to values reported for natural language, and what does this imply about structured domains? **Q3:** Does Maximal Update Parameterization (μ P) [Yang et al., 2022] enable zero-shot learning rate transfer across model sizes for SVG, as it does for NLP?

1.2 Contributions

Our main contributions are:

1. A scaling-law study on SVG code generation across five decoder-only Transformer size classes (1.3M–88.1M parameters under SP; 1.8M–91.2M under μ P) under both standard parameterization (SP) and μ P; for μ P only, the five sizes follow a power-law fit $\alpha = 0.043 \pm 0.715$, $c \approx 0$ ($c_{\text{err}} = 7.10$), $R^2 = 0.76$, as SP exhibits divergence at larger sizes precluding a meaningful fit. The large standard errors mean the fit captures the general downward trend but cannot precisely quantify the scaling exponent or irreducible loss.
2. A direct comparison showing that SP’s Tiny-optimal learning rate causes Large and XL models to diverge, while μ P enables zero-shot LR transfer across all five sizes—a prerequisite for meaningful scaling-law fitting.
3. The observation that downstream generation quality (completion rate, render success) does not track loss: quality peaks at Medium (13.8M parameters, 90% unconditional render rate) while loss continues decreasing at larger scales, suggesting that scaling laws predict loss trends but not practical output quality.

2 Data and Preprocessing

2.1 Dataset Sources

We use two SVG datasets from the StarVector collection [Rodriguez et al., 2023]: `svg-icons-simple` (89,370 records, primary) and `svg-emoji-simple` (5,106 records, supplementary). The emoji dataset was added because the icons-only corpus yielded 98.1M training tokens after tokenization, falling below the project’s 100M-token minimum. The combined corpus contains 94,476 records.

2.2 Preprocessing Pipeline

Both datasets were processed through an identical pipeline: comment and metadata removal, coordinate normalization (1-decimal-place precision), whitespace compression, CairoSVG render verification, and a minimum character-length filter (≥ 50 characters). All 94,476 records passed every stage with zero rejections (Table 1). **Deviation from specification:** the project brief calls for a maximum-length filter to remove long SVGs; however, applying a 1,024-token threshold would retain only 55.7% of sequences, reducing the corpus to ≈ 30 M tokens—well below the 100M-token minimum. We therefore omitted the upper-bound filter to satisfy the data-quantity requirement, accepting that 44.3% of tokenized sequences exceed the 1,024-token context window and span multiple windows during training (Section 5.7). This represents a trade-off between two project requirements rather than a standard filtering decision.

Table 1: Dataset filtering pipeline statistics. Records removed at each stage of the cleaning pipeline. The maximum-length filter specified in the project brief was omitted to meet the 100M-token requirement (see text); this deviation is discussed in Section 5.7.

Stage	Records	Removed
Initial (svg-icons-simple + svg-emoji-simple)	94,476	–
After short-length filter (≥ 50 chars)	94,476	0
After XML parse validation	94,476	0
After render validation	94,476	0
Token-length upper-bound filter	(disabled)	0
Final cleaned corpus	94,476	–

2.3 Tokenization

We trained a byte-level BPE tokenizer (HuggingFace `tokenizers` library) on the combined training split (92,585 sequences) with a target vocabulary size of 4,096. Due to coordinate normalization reducing character diversity to approximately 88 distinct types, BPE merges exhausted after 117 merges, yielding an effective vocabulary of 166 tokens. The implications of this effective vocabulary reduction are discussed in Section 5.1. Figure 1 shows the resulting token-length distribution.

Table 2: Dataset split statistics. Records after filtering.

Split	Records	Percentage
Train	92,585	98.0%
Validation	945	1.0%
Test	946	1.0%
Total	94,476	100%

2.4 Final Corpus Statistics

After tokenization, the training split contains 107M tokens, satisfying the 100M-token project requirement. Tables 2 and 3 summarize the split-level statistics.

Table 3: Token counts by split after tokenization with the 166-token BPE vocabulary.

Split	Tokens
Train	107M
Validation	1.13M
Test	1.10M

The train-split token-length distribution (Figure 1) has median 919, mean 1,157, and p95 2,768. Only 55.7% of sequences fit within the 1,024-token context window; the remainder span multiple windows (Section 5.7). Figure 2 illustrates the visual diversity of the dataset.

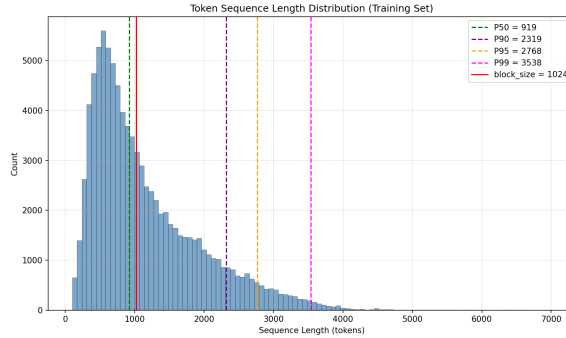


Figure 1: Token length distribution on the train split (92,585 sequences). Median 919, mean 1,157, p95 2,768. Dashed line: 1,024-token context window (55.7% fit).



Figure 2: Rendered SVG examples from the dataset, illustrating the range of complexity from simple icons (left) to detailed emoji (right).

3 Methods

3.1 Model Architectures

We use a decoder-only Transformer architecture derived from nanoGPT [Karpathy, 2022]. We borrowed the core model structure and training loop scaffold, and modified it by removing GPT-2 pretrained weight loading, adding an explicit d_{ff} configuration, setting `bias=False` as default, and integrating μP support via the `mup` package (MuReadout, MuAdamW, `set_base_shapes`). Under our μP implementation, we untie weights between the input embedding and output projection because they follow different scaling rules under μP : the input embedding behaves as a *vector-like* parameter while the output projection behaves as a *matrix-like* parameter, in the terminology of Yang et al. [2022]. Table 4 lists the five model configurations from Tiny (1.8M) to XL (91.2M parameters under μP ; SP counts differ due to weight tying—see footnote).

Table 4: Model configurations and training statistics. μ P unties weight between input embedding and output projection.

Name	d_{model}	n_{layer}	n_{head}	d_{ff}	Params ¹	Time (min)	Mem (GB)	Tok/s
Tiny	128	4	4	512	1.8M	2.7	1.6	785k
Small	192	6	6	768	4.2M	5.6	2.3	375k
Medium	384	6	6	1536	13.8M	12.5	3.5	167k
Large	512	10	8	2048	35.7M	31.5	6.5	69k
XL	768	12	12	3072	91.2M	70.0	6.4	28k

3.2 Training Setup

All models are trained with AdamW [Loshchilov and Hutter, 2019] ($\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.1, gradient clipping at 1.0) using a cosine learning rate schedule with 3% warmup, decaying to $\text{max_lr}/10$. For the five-size scaling runs, training runs for 1 epoch (Kaplan-style without-replacement iteration). The effective batch size is 16,384 tokens across all models: Tiny–Large use 16 sequences $\times$ 1,024-token context with no gradient accumulation, while XL uses 8 sequences $\times$ 2 accumulation steps. Training documents are concatenated into a single token stream with BOS/EOS delimiters; the stream is split into non-overlapping 1,025-token windows that are shuffled and iterated without replacement. Dropout is set to 0.0 for all models. All training was performed on Google Colab Pro with an A100 40GB GPU. All SP and μ P runs complete at step 6,533 (one epoch over 107M tokens with effective batch size 16,384 tokens); validation is evaluated every 200 steps.

3.3 μ P Implementation

We implement μ P [Yang et al., 2022] using the mup package with a base width of 128 (Tiny model). The key modifications are: width-dependent initialization scaling, per-layer learning rate multipliers, and $1/d$ attention scaling (replacing the standard $1/\sqrt{d}$). These changes ensure that activation magnitudes remain stable as model width increases. We verify the implementation via a coord check (Appendix C), confirming that μ P activations are width-independent while SP activations drift with width.

3.4 Experimental Design

The study follows a two-phase design. In Phase 1 (SP baseline), we sweep 5 learning rates on the Tiny model on a log scale (3×10^{-5} to 3×10^{-3}), select the optimal, and train all 5 sizes at that rate. In Phase 2 (μ P), we repeat with 7 learning rates (3×10^{-5} to 3×10^{-2}), again selecting the Tiny-optimal and applying it across all sizes. We fit a Kaplan-style N -only power law

$$L(N) = a \cdot N^{-\alpha} + c \quad (1)$$

because our experimental design fixes $D = 107\text{M}$ tokens across all sizes; the Chinchilla joint $L(N, D)$ form [Hoffmann et al., 2022] requires varying D , which is outside our scope. We separately use Chinchilla’s $D/N \approx 20$ heuristic in Section 5.3 to interpret training regime, but do not fit the Chinchilla law itself. The μ P scaling-law fit is then extrapolated to $10\times$ the largest model size. Beyond these five-size scaling runs, the μ P XL configuration is additionally trained for 2 epochs (extended training); this run serves as the best-model checkpoint for generation evaluation in Section 4.8.

Reproducibility: final artifact mapping. All numerical results reported in this paper are derived from the following runs in the repository: `results/runs/sp/` (SP scaling, 5 sizes), `results/runs/mup/` (μ P scaling, 5 sizes), and `results/runs/mup_xl_extended/` (μ P XL extended training). Evaluation metrics come from `results/evaluation_v2/`. The repository also contains earlier pilot runs under `results/runs/scaling_study/` and `results/runs/mup_scaling_study/`; these used a different configuration and are *not* the source for any reported numbers.

¹Parameter counts shown are for μ P configurations (no weight tying between `lm_head` and `wte`). SP uses weight tying, reducing counts by $V_{\text{config}} \times d_{\text{model}}$: SP Tiny 1.3M vs. μ P 1.8M; Small 3.4M vs. 4.2M; Medium 12.2M vs. 13.8M; Large 33.6M vs. 35.7M; XL 88.1M vs. 91.2M (source: `summary.json`).

3.5 Evaluation Metrics

Training metrics are cross-entropy loss and perplexity on non-overlapping validation/test windows. Generation metrics on sampled SVGs are: *completion rate* (`</svg>` within 4,096 tokens), *XML validity*, *structural validity* (valid root, closed tags, parseable attributes), and *render rate* (CairoSVG success). In practice, these collapse to two independent signals—completion and render—as structural/XML validity always coincide with render success in our data. Sampling uses top- k and top- p with $t \in \{0.5, 0.8, 1.0\}$.

4 Results

4.1 SP Learning Rate Sweep

We swept 5 learning rates on a log scale (3×10^{-5} , 1×10^{-4} , 3×10^{-4} , 1×10^{-3} , 3×10^{-3}) for the SP Tiny model. As shown in Figure 3, 3×10^{-3} achieves the lowest validation loss and is selected as the optimal SP learning rate for all subsequent sizes.

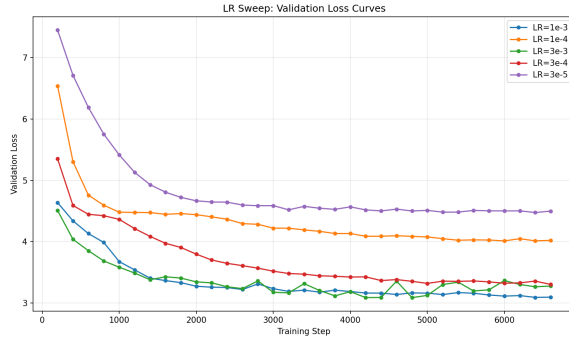


Figure 3: SP learning rate sweep on the Tiny model. Optimal LR = 3×10^{-3} minimizes validation loss.

4.2 SP Scaling Study

Applying the Tiny-optimal LR (3×10^{-3}) to all five sizes, Tiny through Medium converge stably with final validation losses of 0.4796, 0.4071, and 0.4013, respectively (Table 5). However, Large and XL diverge, ending at 1.305 and 1.667—losses above their early-training values. This divergence is visible in the training curves (Figure 7, left column). Fitting the power law to SP yields $\alpha = 1.76$, $c = 0.85$, $R^2 < 10^{-10}$ (Section 4.4), but the fit is dominated by the diverged Large/XL points and has no predictive value—we treat this as evidence that SP’s instability at scale precludes a meaningful scaling law, providing an empirical motivation for μP .

Table 5: SP scaling study results: validation loss after 1 epoch with optimal Tiny LR (3×10^{-3}) applied to all sizes. Parameter counts reflect SP (with weight tying); see Table 4 footnote for μP counts.

Name	Params	Final validation loss	Status
Tiny	1.3M	0.4796	converged
Small	3.4M	0.4071	converged
Medium	12.2M	0.4013	converged
Large	33.6M	1.305	diverged
XL	88.1M	1.667	diverged

4.3 μP Learning Rate Sweep

Under μP , we swept 7 learning rates (3×10^{-5} to 3×10^{-2}) on the Tiny model. The optimal LR shifts from 3×10^{-3} (SP) to 1×10^{-2} (μP), a roughly $3 \times$ increase (Figure 4). This higher optimal LR is consistent with μP ’s width-independent gradient scaling.

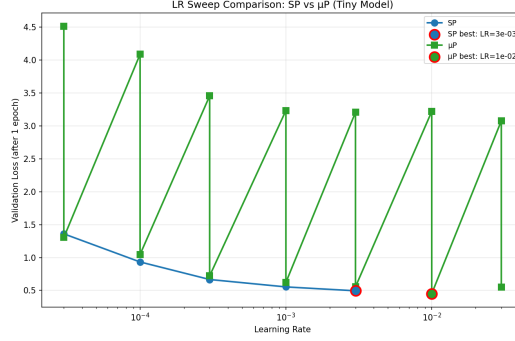


Figure 4: Learning rate sweep comparison: SP (left) vs μ P (right) on the Tiny model. Optimal LR shifts from 3×10^{-3} (SP) to 1×10^{-2} (μ P).

4.4 μ P Scaling Study

Applying the μ P Tiny-optimal LR (1×10^{-2}) to all five sizes, all five models converge, with final validation losses ranging from 0.4712 (Tiny) to 0.3807 (XL). Table 6 reports these values alongside the SP contrast; the Δ column shows that μ P and SP perform comparably at Tiny–Medium ($|\Delta| \leq 0.031$, with mixed signs), while μ P dramatically outperforms at Large and XL where SP diverges ($\Delta < -0.85$).

We fit $L(N) = a \cdot N^{-\alpha} + c$ (Equation 1) to the five μ P data points, obtaining $a = 0.85$, $\alpha = 0.043$, $c \approx 0$, $R^2 = 0.76$ (Figure 5). The fit carries large standard errors ($\alpha_{\text{err}} = 0.715$, $c_{\text{err}} = 7.10$), reflecting the limitation of 3 parameters fit to 5 points. The observed per-size losses are non-monotonic (Small $\rightarrow$ Medium: +0.013), though with $n=5$ sizes this bump is indistinguishable from single-seed variability (Sections 5.4, 5.7). Fitting the same model to SP yields $a = 1.07$, $\alpha = 1.76$, $c = 0.85$, $R^2 < 10^{-10}$ (effectively zero)—the diverged Large/XL points dominate the fit, producing a nonsensical exponent with no predictive value. The inability to extract a meaningful scaling law from SP is evidence that, under this fixed-LR transfer protocol, μ P provides a clear scaling advantage; whether this generalizes to other LR schedules (e.g., per-size tuning) remains untested.

The μ P XL model was additionally trained for $2 \times$ steps (extended training), reaching a best-checkpoint val loss of 0.3506 (ppl 1.42; final val loss 0.3562), and serves as the best-model configuration for generation evaluation (Section 4.8).

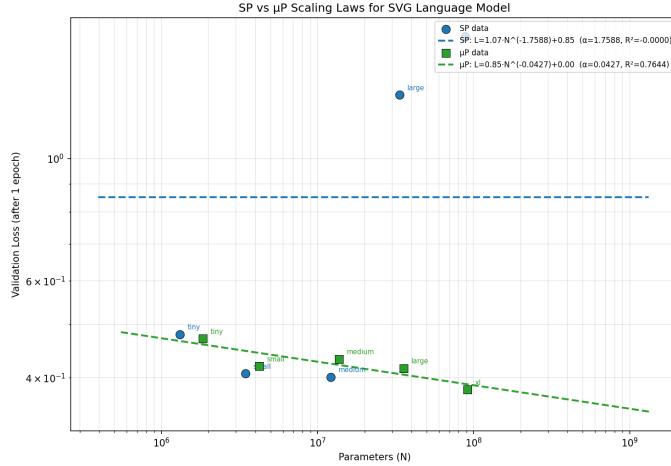


Figure 5: Scaling curves under SP (red) and μ P (blue). X-axis: parameter counts under each parameterization (SP uses weight tying; μ P unties `wte/lm_head`—see Table 4 footnote and Tables 5–6). SP Large/XL diverge from the fitted curve (no meaningful power law); μ P follows $L = 0.85 \cdot N^{-0.043}$ with $R^2 = 0.76$.

²Best-checkpoint validation loss (lowest during training); final validation loss at step 13,066 was 0.3562. All generation evaluation uses the best checkpoint.

Table 6: μP scaling study results with SP contrast. The upper section (Part 3: scaling-law comparison) reports 5 sizes after one epoch under fixed Tiny-optimal LR. The lower section (Part 4: best-model evaluation) reports the μP XL model after extended training ($2\times$ steps) used for generation evaluation.

Name	Params (μP)	μP final val. loss	SP final val. loss	Δ ($\mu\text{P} - \text{SP}$)
<i>Part 3: 5-size scaling comparison (1 epoch each)</i>				
Tiny	1.8M	0.4712	0.4796	-0.008
Small	4.2M	0.4198	0.4071	+0.013
Medium	13.8M	0.4324	0.4013	+0.031
Large	35.7M	0.4161	1.305	-0.889
XL	91.2M	0.3807	1.667	-1.286
<i>Part 4: best-model extended training</i>				
XL ext. ²	91.2M	0.3506	—	—

4.5 Learning Rate Transfer

Figure 6 summarizes: SP’s Tiny-optimal LR (3×10^{-3}) causes divergence at larger scales, while μP ’s (1×10^{-2}) transfers to all five sizes, confirming zero-shot LR transfer for SVG.

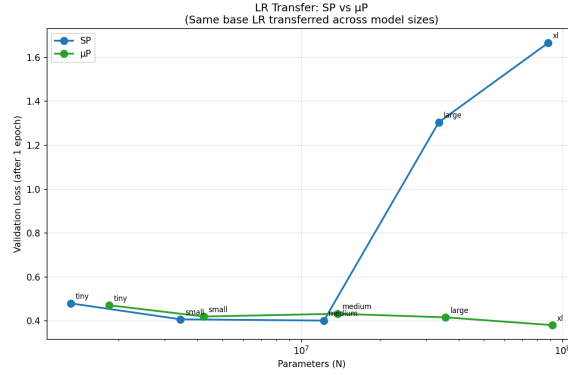


Figure 6: Learning rate transfer: SP requires per-size LR retuning; μP enables zero-shot LR transfer across all 5 model sizes.

4.6 Training Curves

Figure 7 shows loss curves for all 10 models. SP Large/XL exhibit a double-hump divergence pattern (loss rising above 1.0); all five μP models converge stably to 0.38–0.48.

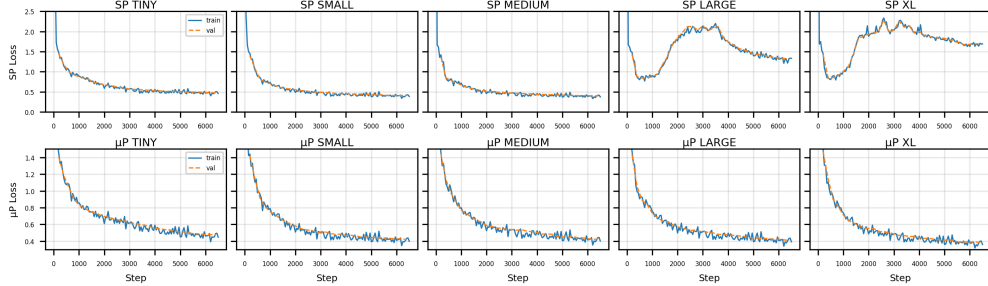


Figure 7: Training and validation loss curves (top: SP, bottom: μP ; left-to-right: Tiny–XL). All runs complete at step 6,533 (one epoch). Final validation losses in Tables 5–6 reflect step 6,533, not the last logged eval point (step 6,400).

4.7 Extrapolation

Extrapolating the μP fit to $10\times$ the largest model size (912M parameters) predicts a validation loss of 0.353, 95% CI [0.159, 0.547]. The CI is computed via Jacobian-based error propagation from the parameter covariance matrix ($n=5$ points, 2 degrees of freedom). The wide CI reflects $\alpha_{\text{err}} = 0.715 \gg \alpha = 0.043$; the extrapolation should be interpreted cautiously (Section 5.7).

4.8 Generation Quality Across Model Sizes

We evaluate generation quality in two analyses. The μ P XL extended model (best-checkpoint val loss 0.3506, test perplexity 1.4249) is used for sampling strategy comparison (Table 9); the train-val gap of 0.0048 at the final step is consistent with no strong overfitting despite 2 epochs. For the five-size scaling comparison, each size uses its best checkpoint from the 1-epoch run (Tables 7, 8).

Per model size, we generate $n=10$ unconditional and $n=15$ prefix-conditioned samples (top- $k=40$, $t=0.8$). Medium achieves the highest unconditional render rate (90%), while smaller and larger models score 60–70%; prefix conditioning shows a similar pattern (Section 5.6). On the XL extended model, prefix conditioning yields +27pp completion (87% vs. 60%) but -7 pp render rate (53% vs. 60%) vs. matched unconditional, with Wilson CIs overlapping for both metrics (Appendix B.3, Table 12). Top- k outperforms top- p at $t=0.8$ (90% vs. 60%); both achieve 100% at $t=1.0$ and $\leq 10\%$ at $t=0.5$ (Table 9). Qualitatively, successful samples exhibit approximate bilateral symmetry and well-formed viewBox declarations; $t=0.8$ balances diversity and structural validity. Figure 8 shows representative unconditional samples; prefix completion examples and cross-size grids are in Appendix B.2–B.1.

Table 7: Unconditional generation metrics across all 5 μ P model sizes ($n=10$ each, decoding: top- $k=40$, $t=0.8$).

Model	n	Complete (%)	XML (%)	Struct (%)	Render (%)
Tiny	10	70	70	70	70
Small	10	60	60	60	60
Medium	10	90	90	90	90
Large	10	60	60	60	60
XL	10	60	60	60	60

[†]95% Wilson CI ($n=10$; completion rate = render rate for unconditional samples):

Tiny [40–89%], Small [31–83%], Medium [60–98%], Large [31–83%], XL [31–83%].

Table 8: Prefix-conditioned generation metrics across all 5 μ P model sizes ($n=15$ each: 5 face_partial + 5 open_path + 5 single_shape_group; decoding: top- $k=40$, $t=0.8$).

Model	n	Complete (%)	XML (%)	Struct (%)	Render (%)
Tiny	15	80	33	33	33
Small	15	73	47	47	47
Medium	15	100	67	67	67
Large	15	60	40	40	40
XL	15	80	20	20	20

[†]95% Wilson CI for completion rate ($n=15$):

Tiny [55–93%], Small [48–89%], Medium [80–100%], Large [36–80%], XL [55–93%].

[‡]95% Wilson CI for render rate ($n=15$):

Tiny [15–58%], Small [25–70%], Medium [42–85%], Large [20–64%], XL [7–45%].

Table 9: Sampling strategy comparison on the μ P XL extended model ($n=10$ unconditional samples per condition; Part 4 best-model evaluation).

Sampling	Temp	Complete (%)	XML (%)	Struct (%)	Render (%)
top- $k=40$	0.5	10	10	10	10
top- $k=40$	0.8	90	90	90	90
top- $k=40$	1.0	100	100	100	100
top- $p=0.95$	0.5	0	0	0	0
top- $p=0.95$	0.8	60	60	60	60
top- $p=0.95$	1.0	100	100	100	100

5 Discussion

5.1 Tokenization Strategy: Effective Vocabulary Reduction

We targeted a BPE vocabulary of 4,096 tokens, but coordinate normalization (1-decimal-place rounding) reduced effective character diversity to approximately 88 types, so BPE merges exhausted after only 117 operations, yielding 166 tokens. This reflects SVG’s structural regularity: the token



Figure 8: Five representative samples from the μP XL extended model (top- p $p=0.95$, $t=0.8$; corresponds to Table 9 top- p row and Table 12 unconditional row, $n=10$, render rate 60%). A full cross-size comparison grid (top- $k=40$, $t=0.8$; Table 7) is provided in Appendix B.1.

space is dominated by XML tags, attribute names, and normalized numerics. Per-token entropy is consequently lower than in natural language, partly explaining the low absolute losses across all sizes.

Parameter count implications. The model allocates embedding (wte) and output-projection (lm_head) matrices for the full configured vocabulary ($V_{\text{config}} = 4,096$), so 3,930 unused positions occupy parameters that do not contribute to SVG modeling. Under μP (no weight tying), the unused wte rows receive zero gradients (0.5M for Tiny, 28% of N ; 3.0M for XL, 3.3%); the unused lm_head rows receive gradients via the full-vocabulary softmax but learn only to suppress their logits—a trivial task. The vocabulary-related parameters thus constitute 58% of Tiny’s N but only 7% of XL’s, making the effective capacity range wider than the reported $50\times$ ratio. This non-uniform distortion could bias α and inflate D/N ratios for smaller models (Tiny’s effective $D/N \approx 82\times$ when computed against vocabulary-adjusted parameters [$N_{\text{eff}} = 1.3\text{M}$, excluding 0.5M unused embedding rows] vs. reported $60\times$ in Table 10 against the full μP parameter count [1.8M]; XL’s effective $D/N \approx 1.2\times$ is essentially unchanged because the unused vocab is only 3.3% of XL’s parameters). We report total parameter counts for reproducibility, but this caveat should be noted when interpreting the scaling fit.

5.2 Architecture Choices

We use a decoder-only architecture spanning 1.8M–91.2M parameters by scaling both width (d_{model} from 128 to 768) and depth (n_{layer} from 4 to 12), following width–depth ratios used in prior scaling-law studies. The context window is set to 1,024 tokens: self-attention cost scales as $O(T^2)$, so doubling to 2,048 would quadruple attention compute per step and exceed our single-GPU time budget. This is a deliberate trade-off—44.3% of tokenized SVGs exceed 1,024 tokens (Section 5.7)—but the concatenated-stream training scheme ensures that no training data is discarded; longer SVGs simply span multiple context windows.

5.3 Training Decisions and the Chinchilla Trade-off

Table 10 presents the Chinchilla D/N ratio for each size [Hoffmann et al., 2022]. At the optimal $D/N \approx 20$, only Tiny ($60\times$) and Small ($25\times$) meet the recommended allocation; XL operates at $D/N = 1.2\times$ (6% of optimal). This fixed-data design is necessary for valid scaling-law fitting but means larger models are undertrained relative to capacity. Note that “near-optimal” refers to D/N , not loss: the lowest validation loss is achieved by XL (0.3807), not Small (0.4198).

Table 10: Chinchilla token-to-parameter ratio analysis. Params (N) column uses μP parameter counts (Table 4); SP counts would shift Tiny’s ratio from $60\times$ to $82\times$ (see Section 5.1 for vocabulary-adjusted ratios). The Chinchilla-optimal ratio is approximately $D/N \approx 20$.

Model	Params (N)	Tokens (D)	D/N	Regime
Tiny	1.8M	107M	$60\times$	Compute-rich ($D/N > 20$)
Small	4.2M	107M	$25\times$	Near-optimal
Medium	13.8M	107M	$7.7\times$	Data-limited
Large	35.7M	107M	$3.0\times$	Severely data-limited
XL	91.2M	107M	$1.2\times$	Severely data-limited

5.4 Scaling Insights: SVG vs NLP α

Our μP fit yields $\alpha = 0.043 \pm 0.715$ and $c \approx 0$ ($c_{\text{err}} = 7.10$, $R^2 = 0.76$). The standard errors substantially exceed both point estimates, so we cannot statistically distinguish α_{SVG} from values reported for natural language (e.g., Kaplan et al. 2020 report $\alpha \approx 0.076$), nor can we conclude that c is meaningfully near zero. With $n=5$ size points and a single seed per size, our fit is moderate in goodness ($R^2 = 0.76$) but underpowered for inter-domain comparison. Qualitatively, the low fitted

α is consistent with the data-limited regime identified in Section 5.3 ($D/N = 1.2\times$ for XL): when models are starved of data, increasing N yields diminishing returns, flattening the apparent scaling curve. Whether SVG’s structure intrinsically yields a different α than NLP cannot be determined from this experiment, as the data-limited confound is sufficient to explain the observed value.

Beyond α , our fit yields $c \approx 0$ with $c_{\text{err}} = 7.10$, meaning c could plausibly fall anywhere in $[-14, +14]$. Kaplan et al. [2020] report a non-zero irreducible loss for natural language (their Eq. 1.5 fits include a positive c). Our data cannot distinguish whether SVG has a meaningfully smaller irreducible loss than NLP—the standard error swamps any signal—but the point estimate is consistent with the hypothesis that SVG’s deterministic syntax leaves less residual entropy than natural language. This question warrants follow-up with more size points and multiple seeds.

5.5 Learning Rate Scaling: SP vs μ P

SP’s Tiny-optimal LR (3×10^{-3}) causes Large/XL divergence, while μ P’s (1×10^{-2}) transfers to all five sizes without adjustment—confirming zero-shot LR transfer for SVG and enabling the scaling-law analysis itself.

5.6 Generation Quality vs. Scaling: A Non-Monotonic Relationship

While the μ P scaling law predicts monotonic loss reduction ($R^2 = 0.76$), generation quality does not track loss: unconditional render rates peak at Medium (90%) while smaller (60–70%) and larger models (60%) score lower; prefix conditioning shows a similar pattern. With small samples ($n=10/n=15$), CIs are wide and this pattern should be treated as a hypothesis rather than a confirmed finding. One plausible explanation is the Chinchilla-suboptimal regime: smaller models lack capacity while larger models are data-starved ($D/N = 3.0\times$ and $1.2\times$), with Medium at a favorable intermediate point. However, this alone cannot explain why Tiny ($D/N=60\times$, compute-rich) achieves higher render rate than Small ($D/N=25\times$, near-optimal). Alternative hypotheses include: (1) generation-length mismatch—larger models may produce longer, more error-prone sequences before EOS; (2) mode collapse at Tiny producing simpler but valid outputs; (3) single-seed variance dominating the $n=10$ sample size. Disentangling these requires multiple seeds and length-stratified analysis, which we leave to future work.

5.7 Limitations and Challenges

Several limitations constrain our findings. (1) Single-epoch, single-seed training: per-size variance is unestimable. (2) The scaling fit ($n=5$, 3 parameters) has $\alpha_{\text{err}} = 0.715 \gg \alpha = 0.043$; domain comparison with NLP is statistically unsupported and the $10\times$ extrapolation CI is wide. (3) Generation samples are small ($n=10/n=15$); the non-monotonic quality pattern is qualitative. (4) 44.3% of sequences exceed the 1,024-token context window and span multiple windows during training (cross-window rate), creating a train–generation mismatch: the model sees these SVGs only in fragments, never as complete sequences. (5) The effective vocabulary (166 vs. 4,096 configured tokens) inflates parameter counts by up to 58% for Tiny, distorting scaling-law inputs non-uniformly across sizes (Section 5.1).

6 Conclusion

Q1: We cannot conclusively answer whether a power law holds for SVG with $n=5$ points and a single seed per size. The μ P validation losses do decrease with N ($R^2 = 0.76$ for the fitted form), but the standard error on α (0.715) exceeds the point estimate (0.043) by an order of magnitude, so we cannot statistically distinguish a power law from alternative monotonic forms. We report $\alpha = 0.043 \pm 0.715$ for completeness. **Q2:** The low α is consistent with the data-limited regime (Section 5.3) rather than an intrinsic SVG property; standard errors preclude comparison with NLP. **Q3:** μ P enables zero-shot LR transfer across all five sizes; SP’s Tiny-optimal LR causes Large/XL divergence.

Two SVG-specific findings emerged: (1) coordinate normalization yielded only 166 effective BPE tokens (vs. 4,096 configured), complicating NLP comparisons, and (2) generation quality does not clearly track loss—Medium shows the highest render rate (90%, $n=10$) while models at other sizes score 60–70% (CIs overlap with Medium’s [60–98%]), consistent with the Chinchilla data-limited regime. Future work should add unique SVG data to move Large/XL toward Chinchilla-optimal D/N , and increase to $n \geq 10$ sizes with multiple seeds.

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A Deliverable Mapping

Table 11 maps each project specification deliverable to the corresponding section, table, or figure in this report.

Table 11: Mapping from project specification deliverables (Parts 1–5) to report sections, tables, and figures.

Deliverable	Location
Part 1: Data	
Filtering pipeline (max-length omitted; see §2.2)	§2.2, Table 1
Tokenization (BPE)	§2.3
Train/val/test splits	§2.4, Table 2
100M-token corpus	§2.4, Table 3
Rendered SVG examples	§2.4, Figure 2
Part 2: SP Scaling	
LR sweep (5 LRs)	§4.1, Figure 3
SP scaling table	§4.2, Table 5
Training curves	§4.6, Figure 7
Time/Mem/Tok/s	Table 4
Part 3: μP	
LR sweep (7 LRs)	§4.3, Figure 4
SP vs μP comparison	§4.4, Table 6, Figure 5
Power-law fit	§4.4, Eq. 1
LR transfer	§4.5, Figure 6
$10\times$ extrapolation + CI	§4.7
Part 4: Best Model + Generation	
XL extended model	§4.4 (last paragraph), §4.8
10 unconditional + 15 prefix (5 per type)	§4.8, Tables 7–8
Temperature {0.5, 0.8, 1.0}	§4.8, Table 9
top- k & top- p	§4.8, Table 9
XML/render/structural metrics	§4.8, Tables 7–9
Test perplexity	§4.8 (text)
Prefix/completion/rendered triplet	§4.8, Figure 10 (Appendix B.2)
Qualitative analysis	§4.8, Figure 8; Appendix B.1–B.3
Part 5: Design Decisions	
Tokenization rationale	§5.1
Architecture rationale	§5.2
Training decisions (Chinchilla)	§5.3, Table 10
Scaling insights (SVG vs NLP)	§5.4
LR insights (SP vs μP)	§5.5
SVG-specific patterns	§5.6
Challenges/limitations	§5.7

B Additional Generation Samples

B.1 Unconditional Samples Across Sizes

Figure 9 extends the representative XL samples shown in the main text (Figure 8) to all 5 model sizes.

B.2 Prefix Completion Across Sizes

Figure 10 shows prefix completion from the XL extended model with prefix text, model completion text, and rendered result. Figure 11 provides a render-only cross-size overview across all 5 sizes.

B.3 XL Extended Matched Comparison

Section 4.8 reports a +27pp completion gain and -7 pp XML validity decrease for prefix conditioning relative to matched unconditional samples on the μP XL extended model. Table 12 provides the underlying per-metric breakdown.

TINY (1.8M, loss=0.471)



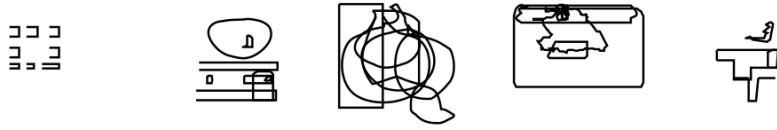
SMALL (4.2M, loss=0.417)



MEDIUM (13.8M, loss=0.423)



LARGE (35.7M, loss=0.410)



XL (91.2M, loss=0.381)

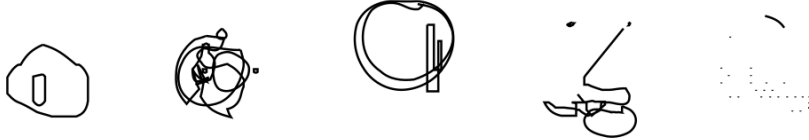


Figure 9: Unconditional generation samples across all 5 μP model sizes (10 samples each, decoding: top- $k=40$, $t=0.8$). Aligned with Table 7. Main text Figure 8 shows the XL row only.

Table 12: Matched comparison on the μP XL extended model: prefix-conditioned ($n=15$: 5 face_partial + 5 open_path + 5 single_shape_group) vs. unconditional ($n=10$) under identical decoding (top- $p=0.95$, $t=0.8$). Source: prefix_vs_unconditional_matched.json.

Condition	n	Complete (%)	XML (%)	Struct (%)	Render (%)
Unconditional	10	60	60	60	60
Prefix-conditioned	15	87	53	53	53
Δ (Prefix – Unconditional)	—	+27	−7	−7	−7

[†]95% Wilson CI: prefix completion ($n=15$, $k=13$) [62–96%];

unconditional completion ($n=10$, $k=6$) [31–83%]; CIs overlap, Δ qualitative only.

C Coord Check Verification

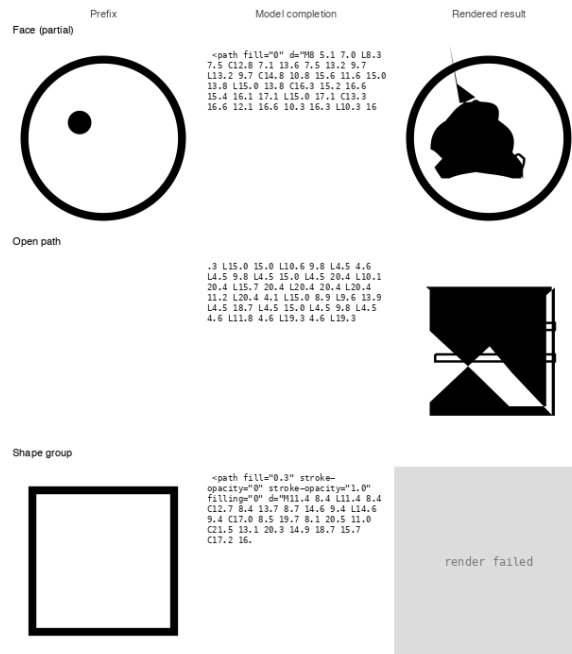


Figure 10: Prefix completion from the μP XL extended model (top- p $p=0.95$, $t=0.8$). Left: prefix. Center: completion text. Right: rendered result. `single_shape_group` failed to render (0/5).

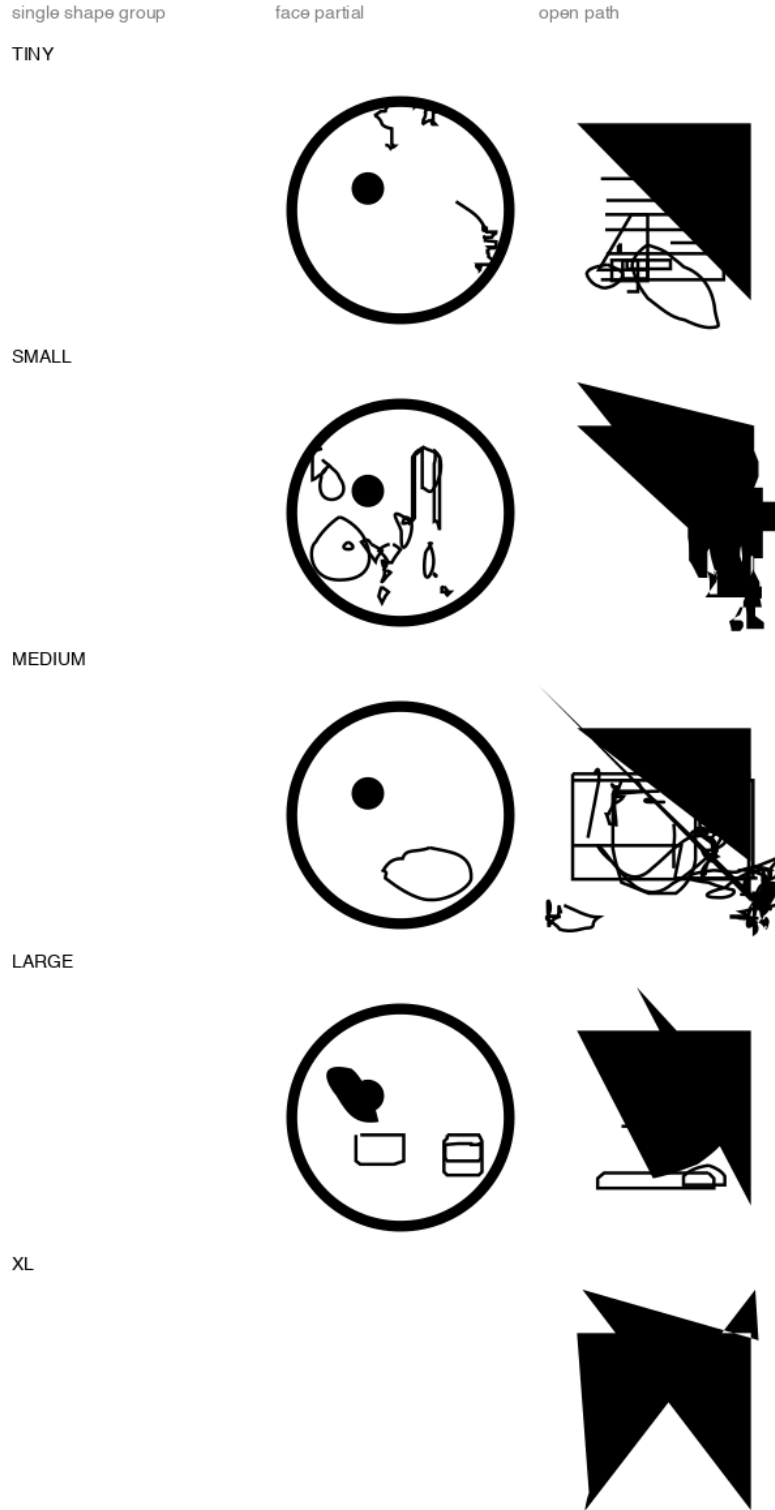


Figure 11: Rendered completions across all 5 μ P model sizes (render-only; see main text Figure 10 for prefix/completion/rendered triplet for XL). 15 samples each: 5 face_partial + 5 open_path + 5 single_shape_group. Aligned with Table 8.

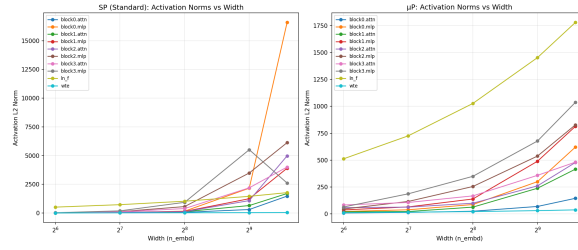


Figure 12: Coord check verification of μP implementation. Activation magnitudes remain stable across model widths under μP , in contrast to SP which exhibits width-dependent drift.